

Extending Permutation Arrays for Prime Powers

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Abstract

Let $M(n, d)$ be the maximum number of permutations of length n with pairwise Hamming distance at least d . We use the technique of partition and extension on $AGL(1, q)$, the *affine general linear group*, for $q = p^k$, where p is a prime and $k > 1$, to obtain improved lower bounds for $M(q + 1, q)$. We also show that, for all prime powers q , $M(q + 2, q) \geq q(q - 1)$.

1 Introduction

In 2000, Han Vinck [22] proposed a specific modulation scheme using permutation codes (or permutation arrays). The error-correcting capability of a permutation code is related to its minimum Hamming distance. Motivated by applications in power line communications [13, 17, 22], extensive research has been conducted on permutation arrays using Hamming distance [1–5, 7–9, 12–14, 16, 19, 20].

Chu *et al.* [9] presented permutation arrays constructed using certain families of polynomials over finite fields, automorphism groups, and a recursive construction. Smith and Montemanni [19] presented several techniques including construction by automorphism group and clique search. Abdollahi et al [1] gave improved upper bounds for Hamming distance 5. Barta *et al.* [4] applied a branch and bound approach combining combinatorial techniques with an approach based on group orbits. Janiszczak *et al.* [14, 16] studied permutation codes invariant under isometries. Permutation codes constructed using Kronecker products and tilings were presented in [6]. The technique of *partition and extension* was presented in [8] and used in other papers [5, 7].

Let σ and π be permutations on n symbols. The Hamming distance between σ and π , denoted by $hd(\sigma, \pi)$ is the number of positions, x , such that $\sigma(x) \neq \pi(x)$. A *permutation array (PA)*, A , has Hamming distance d , denoted by $hd(A) = d$, if every pair of distinct permutations in A has Hamming distance at least d . Two PAs A and B have pairwise Hamming distance d , denoted by $hd(A, B) = d$, if for every pair of distinct permutations $\sigma \in A$ and $\pi \in B$, $hd(\sigma, \pi) \geq d$.

We are interested in finding large PAs with Hamming distance d . Let $M(n, d)$ be the maximum number of permutations of length n and pairwise Hamming distance d . It is known that, if q is either prime or a power of a prime, then $M(q, q - 1) = q(q - 1)$ and $M(q + 1, q - 1) = (q + 1)q(q - 1)$ [11, 18]. It is also known that, for any positive integer n , if there are k mutually orthogonal Latin squares of side n , then $M(n, n - 1) \geq kn$ [10].

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In this paper, we use the technique of partition and extension [8] on $AGL(1, q)$, where $q = p^k$, for prime p and $k > 1$, to obtain new lower bounds for $M(q + 1, q)$. In addition, we use a new technique to show that $M(q + 2, q) \geq q(q - 1)$.

In Section 2 we review the technique of partition and extension. In Section 3 we describe partitions $\mathcal{P}$ and $\mathcal{Q}$ for partition and extension to create PAs on $q + 1$ symbols from permutations in $AGL(1, q)$, for $q = p^k$, where p is a prime and $k > 1$ is even. In Section 4 we give constructions for odd powers of a prime. In Section 5 a dynamic programming algorithm is given that improves in many cases the results given in Section 4. The improved results are given in X. In Section 6 we give new results for $M(p^k + 2, p^k)$.

2 Partition and Extension for $AGL(1, q)$

Consider $GF(q)$, where $q = p^k$. Let $P(x)$ be a degree k primitive polynomial used in the definition of the elements of $GF(q)$, so that the elements of $GF(q)$ are $\mathbf{0}, \mathbf{1} = t^0, \mathbf{2} = t, \dots, \mathbf{q} - \mathbf{1} = t^{q-2}$, where t is a generator of the non-zero elements of $GF(q)$. So, the non-zero elements of $GF(q)$ can be represented as $t^i = \mathbf{i} + \mathbf{1}$, for all i , ($0 \leq i \leq q - 2$). Here we use boldface numerals to designate the symbols $\mathbf{0}, \mathbf{1} = t^0, \mathbf{2} = t, \dots, \mathbf{q} - \mathbf{1} = t^{q-2}$. For ease of notation we will often not use bold font for the elements of $GF(q)$. The context should make it clear when numbers represent field elements.

$AGL(1, q)$ is the sharply 2-transitive group of permutations $\{ax + b \mid a, b, x \in GF(q), a \neq 0\}$, with the operation of function composition. It is known that $AGL(1, q)$ is a permutation array with $q(q - 1)$ permutations and Hamming distance $q - 1$. Furthermore, $AGL(1, q)$ can be partitioned into a collection $\mathcal{C}$ of $q - 1$ cosets, $\mathcal{C} = \{C_0, C_1, \dots, C_{q-2}\}$, where C_0 is a subgroup of $AGL(1, q)$. That is, C_0 is the set of permutations $\{x + b \mid x, b \in GF(q)\}$, and the i^{th} coset, denoted by C_i , for all i ($0 \leq i \leq q - 2$), consists of the set of permutations $\{(i + 1)x + b \mid i, x, b \in GF(q)\}$. We view each C_i as an array with q rows and q columns, where the x^{th} column of the b^{th} row, for $0 \leq x \leq q - 1$ and $0 \leq b \leq q - 1$, is the symbol $(i + 1)x + b$. Note that the Hamming distance of each coset is q , and the pairwise Hamming distance between cosets is $q - 1$. In fact, each coset is a Latin square [9, 21].

An element t^i of $GF(q)$ can also be represented as the mod p polynomial remainder of t^i divided by a primitive polynomial $P(t)$. That is, t^i can be written as an k -tuple $(a_1, a_2, \dots, a_k)$, where $(a_1, a_2, \dots, a_k)$ represents the coefficients of the polynomial remainder. Conversely, let $Number(a_1, a_2, \dots, a_k)$ denote the element t^i described by the k -tuple $(a_1, a_2, \dots, a_k)$ and, for a set A of such k -tuples, let $Number(A)$ denote the set of elements in $GF(q)$ represented. The k -tuple $(0, \dots, 0)$ represents the element 0. For convenience, we often refer to the elements of $GF(q)$ as length k strings of symbols, $a_1 a_2 \dots a_k$ rather than as an k -tuple. Addition of the k -tuples (or length k strings) is done separately on each coordinate (or position, respectively) using mod p arithmetic. For example, $(0, 1, 1) + (1, 1, 0) = (1, 0, 1)$ in mod 2 arithmetic.

As an example, let $p = 2$ and $k = 3$, so that $q = 2^3$. Table 1 shows the permutations in C_0 for the group $AGL(1, 2^3)$, using the primitive polynomial $P(x) = x^3 + x + 1$. Observe that there are different ways to describe rows and columns in a table, such as the one shown in Table 1. For example, one can describe a column by *name*, which is the element in the first row of that column, or alternatively, by *number*, where we count columns from left-to-right starting with column number 0. Row 0 of C_0 is the identity permutation, i.e. $x + 0$. Row 1 is the permutation $\pi(x) = x + 1$, and so on.

Now consider cosets of the subgroup C_0 in the group $AGL(1, 8)$. Table 1 also shows the cosets C_1 and C_2 obtained by composing the permutation represented by $2x$ (respectively, $3x$) with each permutation in C_0 .

0	1	2	3	4	5	6	7	0	2	3	4	5	6	7	1	0	3	4	5	6	7	1	2
1	0	4	7	2	6	5	3	1	4	7	2	6	5	3	0	1	7	2	6	5	3	0	4
2	4	0	5	1	3	7	6	2	0	5	1	3	7	6	4	2	5	1	3	7	6	4	0
3	7	5	0	6	2	4	1	3	5	0	6	2	4	1	7	3	0	6	2	4	1	7	5
4	2	1	6	0	7	3	5	4	1	6	0	7	3	5	2	4	6	0	7	3	5	2	1
5	6	3	2	7	0	1	4	5	3	2	7	0	1	4	6	5	2	7	0	1	4	6	3
6	5	7	4	3	1	0	2	6	7	4	3	1	0	2	5	6	4	3	1	0	2	5	7
7	3	6	1	5	4	2	0	7	6	1	5	4	2	0	3	7	1	5	4	2	0	3	6

Table 1: Subgroup C_0 of $AGL(1, 8)$ and Cosets C_1 and C_2 .

Observe that the columns of C_1 can also be obtained from the columns of C_0 by a shifting operation. Specifically, column 0 of C_1 is the same as column 0 of C_0 , but column i , ($1 \leq i \leq 7$) of C_1 is the same as column $(i + 1) \pmod{7}$ of C_0 . A brief explanation of this follows.

Consider the permutation $\pi(x) = x + b$ in C_0 and the permutation $\tau(x) = a \cdot x + b$ where $a = t^1$ in C_1 . Consider the value of $\tau(t^i)$, i.e. $x = t^i$. We have $\tau(t^i) = \pi(t^{i+1}) + b$, which means that the value in the C_1 , corresponding to $2 \cdot x + b$, is the value in C_0 but shifted by 1 column. Also, when $x = 0$, $\tau(0) = b = \pi(0)$. So, in each row of the coset C_1 , the elements are shifted as described from the same row of elements for C_0 . Similarly, Table 1 shows coset C_2 . It is obtained by the same shift operation from C_1 . Similar shift operations would yield cosets $C_3, C_4, \dots, C_6$.

In general, the subgroup C_0 consists of permutations formed by the rows of the addition table of $GF(q)$. That is, the row with index b in C_0 , which corresponds to $x + b$, gives the sum of each element in $GF(q)$ with b .

Define the permutation σ in cycle notation by $\sigma = (0)(1\ 7\ 6\ 5\ 4\ 3\ 2)$. So, $\sigma^{-1} = (0)(1\ 2\ 3\ 4\ 5\ 6\ 7)$. Each coset C_i consists of rows that can be obtained from the rows of C_0 using σ^{-1} . If $A = a_0\ a_1\ \dots\ a_{q-1}$ is a permutation (row) of C_0 , then $\sigma^{-1} \circ A$ is the permutation in the corresponding row of C_1 . For example, the second row of C_0 is the permutation $A = 1\ 0\ 4\ 7\ 2\ 6\ 5\ 3$, and $\sigma^{-1} \circ A = 1\ 4\ 7\ 2\ 6\ 5\ 3\ 0$, which is the second row of C_1 , given in Table 1. Similarly, If $A = a_0\ a_1\ \dots\ a_{q-1}$ is a permutation (row) of C_0 , then $(\sigma^{-1})^i \circ A$ is the permutation in the corresponding row of C_i . Define the function $shift(i, j)$ by: $shift(i, j) = (\sigma(j))^i$,

We use the technique of partition and extension (denoted by $P\&E$) [8] on groups $AGL(1, q)$, for q a power of a prime, in order to obtain a large family of permutations on $q + 1$ symbols with Hamming distance q . We review the definitions and theorems in [8] for the reader's convenience.

Define a triple $\Pi = (\mathcal{M}, \mathcal{P}, \mathcal{Q})$, where $\mathcal{M} = (M_0, M_1, \dots, M_r)$, for some $r > 0$, is a collection of $r + 1$ disjoint subsets of permutations of length q , and where $\mathcal{P} = (P_0, P_1, \dots, P_{r-1})$ and $\mathcal{Q} = (Q_0, Q_1, \dots, Q_{r-1})$ are partitions of $GF(q)$. Using the terminology given in [8], $\mathcal{P}$ is a *partition of positions* and $\mathcal{Q}$ is a *partition of symbols* (or values). The triple Π is called a *distance- d partition system* when for all i, j ($0 \leq i, j \leq m$, $i \neq j$), $hd(M_i) = d$, and $hd(M_i, M_j) \geq d - 1$. In particular for $AGL(1, q)$, the triple $\Pi = (\mathcal{M}, \mathcal{P}, \mathcal{Q})$ is a distance- q partition system, when $\mathcal{M} = (M_0, M_1, \dots, M_r)$ is a set of $r + 1$ distinct cosets of C_0 , for some r , $0 \leq r \leq q$.

Both of the partitions $\mathcal{P}$ and $\mathcal{Q}$ are ordered sets and the pair (P_i, Q_i) is associated with the coset M_i , for all i , ($0 \leq i \leq r - 1$). The coset M_r is dealt with separately. We say that a permutation π is

covered by a set of positions P_i and set of symbols Q_i if there is an element x in P_i such that $\pi(x)$ is an element of Q_i . If all of the permutations in M_i are covered, then we say that the M_i is *well covered* or *fully covered*. If at least one permutation in M_i is covered, then we say that M_i is *covered*. We create new permutations on $q+1$ symbols by modifying the covered permutations on q symbols in M_i as follows. If a permutation π is covered, say at position x , we change it to a new permutation π' on the symbols $\{0, 1, \dots, q\}$ by substituting the symbol q for the symbol $\pi(x)$ in position x and putting the replaced symbol $\pi(x)$ in position q . We apply this extension algorithm to each covered permutation in M_i , creating a new set of extended permutations, $ext(M_i)$, on $q+1$ symbols. The set $ext(M_r)$ is created by appending the symbol q to the end of each permutation in M_r . (We call this coset a *freebie*, because no partitioning or exchange of elements is required.) As none of the sets P_i ($0 \leq i \leq m-1$), include position q , there is no new agreement created by including $ext(M_r)$. The collection of sets $ext(M_i)$, ($0 \leq i \leq r$) forms a new PA, $ext(\Pi)$, of permutations on $q+1$ symbols with Hamming distance q . The process is described in detail in [8], where the following theorem and corollary are established. Theorem 1 gives us the Hamming distance of the PA, $ext(\Pi)$, on $n+1$ symbols created by partition and extension, and Corollary 2 tells us how to compute the number of permutations in $ext(\Pi)$, giving a lower bound on $M(n+1, d)$.

Theorem 1. [8] *Let d be a positive integer. Let $\Pi = (\mathcal{M}, \mathcal{P}, \mathcal{Q})$ be a distance- d partition system for $[0, 1, \dots, n-1]$, with $\mathcal{M} = (M_0, M_1, \dots, M_r)$ for some positive integer r . Let $ext(\Pi)$ be the PA on $[0, 1, \dots, n]$ created by partition and extension. Then, $hd(ext(\Pi)) \geq d$.*

Corollary 2. [8] *Let $\Pi = (\mathcal{M}, \mathcal{P}, \mathcal{Q})$ be a distance- d partition system for $[0, 1, \dots, n-1]$, with $\mathcal{M} = (M_0, M_1, \dots, M_r)$, and $|M_r| = l$.*

1. *If $\mathcal{M}$ is covered (but not well covered), then $|ext(\Pi)| = \sum_{i=0}^r |ext(M_i)|$.*
2. *If $\mathcal{M}$ is well covered by $\mathcal{P} = (P_0, P_1, \dots, P_{r-1})$ and $\mathcal{Q} = (Q_0, Q_1, \dots, Q_{r-1})$, then $|ext(\Pi)| = l + \sum_{i=0}^{r-1} |P_i| \cdot |Q_i|$. Hence,*

$$M(n+1, d) \geq l + \sum_{i=0}^{r-1} |P_i| \cdot |Q_i|.$$

It should be noted that in our use of *P&E* on $AGL(1, q^k)$, every coset has q^k permutations, so M_r , the so-called *freebie*, adds q^k permutations. On the other hand, in Section 4 there may be one coset, say C_{r-1} , that is not fully covered. If s permutations of C_{r-1} are covered, then $|M_{r-1}| = s$. In this case the second part of Corollary 2 can be written as $M(q+1, q) \geq rq^k + s$, as r cosets are fully covered and the partially covered one adds s permutations.

Let us now give specific distance- q partition systems when $q = p^k$, with $k > 1$. We consider two cases, namely when k is even and when k is odd.

Case (1): k is even.

The permutations in $AGL(1, q)$ are permutations of the symbols of $GF(p^{2m})$, for some $m \geq 1$. So, the elements of $GF(p^{2m})$ can be viewed as length $2m$ strings of distinct symbols from $[0, 1, \dots, p-1]$. Any such string can also be viewed as a prefix of length m followed by a suffix of length m . Let $Suffix(\beta)$ be the set of all strings of length $2m$ over $[0, 1, \dots, p-1]$ whose length m suffix is β . There are p^m different suffix sets $Suffix(\beta)$ for the various β .

Let $\Pi = (\mathcal{M}, \mathcal{P}, \mathcal{Q})$ be the distance- d partition system where $\mathcal{M} = (C_0, C_1, \dots, C_{p^m})$ is the collection consisting of C_0 , the subgroup of $AGL(1, q)$, and the first p^m cosets of C_0 , and where

$\mathcal{P} = (P_0, P_1, \dots, P_{p^m-1})$ is a collection of position sets such that, for all i ($0 \leq i \leq p^m-1$), P_i contains exactly one element of $Suffix(\beta)$, for each possible suffix β , and $\mathcal{Q} = (Q_0, Q_1, \dots, Q_{p^m-1})$ is the collection of all distinct p^m suffix sets, say with suffixes in lexicographic order.

Note that the sets P_i are not completely defined, as only a property of the sets, call it *property I*, is given. That is, *property I* is that each set P_i contains one permutation from each suffix set. A complete definition for P_i will be given in Section 3 where we ensure that the sets P_i are disjoint. We shall call a *distance - d* partition system such as $\Pi = (\mathcal{M}, \mathcal{P}, \mathcal{Q})$ in the previous paragraph a *incomplete distance - d* partition system.

We first show, for any *incomplete distance - d* partition system, say Π_I , that C_0 is well-covered. That is, C_0 is fully covered by any sets with the property I. Consider any permutation $\sigma(x) = x + b$ in C_0 . For the first block C_0 in Π_c , the set of values is Q_0 , which is all strings in $Suffix(0)$. So, we must have $Suffix(x) + Suffix(b) = Suffix(0)$. This is true as the positions for x include one string for each possible suffix. That is, there is one such position for x , whose suffix when added to the suffix of b is equal to 0. (A similar proof shows this is true for the set of values $Suffix(\alpha)$, for any α .)

Note that, if position set P and value set Q cover all the rows of C_0 , then they also cover all the rows of C_i , for all $1 \leq i \leq q-1$. That is, the columns of C_0 are the same as the columns of C_i , they are just shifted.

We observe that the so-called *freebie* coset C_m is also well covered [8]. In Section 3 we give a specific *distance - d* partition system using a maximum matching in a constructed bipartite graph.

Case (2): k is odd.

The permutations in $AGL(1, q)$ are permutations of the symbols of $GF(p^{2m+1})$, for some $m \geq 1$. As their length is odd there is no way to partition the permutations into two equal length halves. In an effort to mitigate this problem we consider special sets. That is, for every permutation π , we consider both its suffix of length m , denoted by $Suffix(\pi)$, and the sum modulo p of all the symbols in its prefix of length $m+1$, denoted (with a slight abuse of notation) as $prefix(\pi)$. The permutations are partitioned into disjoint sets described by these parameters. A *special set*, $i - Suffix(\alpha)$, is defined by $\{ \pi \mid prefix(\pi) = i \text{ and } Suffix(\pi) = \alpha \}$. There are p^m different suffixes, each with p^{m+1} permutations. And there are p different prefixes, splitting the suffix sets into a total of p^{m+1} different special sets, each with p^m permutations.

In order to have a roughly equal number of positions and values for covering cosets we use two methods.

Method 1: take the set of values to be an entire suffix set, say $Suffix(\alpha)$, and the set of positions to be one permutation from each suffix set.

Method 2: take the set of values to be one of the special sets and the set of positions be one choice from each special set.

Note that Method 1 uses a set of p^{m+1} values and a set of p^m positions and Method 2 uses a set of p^m values and a set of p^{m+1} positions. There are a total of p^{2m+1} elements in $GF(p^{2m+1})$, which are considered as both positions and values. If just Method 1 were used, all p^{2m+1} values would be utilized to fully cover p^m cosets, with $p^{2m} * (p-1)$ positions unused. If just Method 2 were used, all p^{2m+1} positions would be utilized to fully cover p^m cosets, with $p^{2m} * (p-1)$ values unused. If one uses Method 1 and Method 2 an equal number of times, $2 * p^{m+1} / (p+1) > p^m$ cosets are fully covered.

To see how many cosets are covered with an equal number of uses of Methods 1 and 2, we define S to be the number of times Method 1 is used and T to be the number of times Method 2 is used.

As $S = T$, we have

$$S - T = 0.$$

Each use of Method 1 takes one element for each prefix for every possible suffix. Each use of Method 2 takes one element for each suffix with some prefix. As there are p prefixes for each suffix and p^m different suffixes, we get

$$pS + T \leq p^{m+1}.$$

Adding these together, we get

$$(p+1)S \leq p^{m+1}, \text{ or, equivalently, } S \leq \frac{p^{m+1}}{p+1}.$$

We want to maximize the number of cosets, so let $S = \left\lfloor \frac{p^{m+1}}{p+1} \right\rfloor$. As $S = T$, the total is $U = 2 \left\lfloor \frac{p^{m+1}}{p+1} \right\rfloor$ cosets.

Let $\Pi = (\mathcal{M}, \mathcal{P}, \mathcal{Q})$ be the incomplete distance- d partition system where $\mathcal{M} = (C_0, C_1, \dots, C_{U-1})$ is the collection consisting of C_0 , the subgroup of $AGL(1, q)$, and the first $U-1$ cosets of C_0 , and where $\mathcal{P} = (P_0, P_1, \dots, P_{U-1})$ is a collection of position sets such that, for all i ($0 \leq i \leq \frac{U}{2} - 1$), P_i contains exactly one element of each suffix set and, for all i ($\frac{U}{2} \leq i \leq U-1$), $P_i = Suffix(\alpha_i)$, and $\mathcal{Q} = (Q_0, Q_1, \dots, Q_{U-1})$ is a collection of value sets such that, for all i ($0 \leq i \leq \frac{U}{2} - 1$), $Q_i = Suffix(\alpha_i)$ and, for all i ($\frac{U}{2} \leq i \leq U-1$), Q_i is one element from each Suffix set.

By an argument similar to that in the previous case, we show all U cosets (and the freebie coset) are fully covered by an *incomplete distance - d* partition system.

Suppose the permutation $x + b$ in coset C_0 is to be covered by Method 1. Let the value set be $Suffix(\alpha)$, for some α . Let $suffix(b) = \gamma$. As the position set includes one from every suffix set, we can choose x to be in $Suffix(\alpha - \gamma)$. So, $x + b$ is in $Suffix(\alpha)$.

Similarly, suppose the permutation $x + b$ in coset C_0 is to be covered by Method 2. Let the value set be $i - Suffix(\alpha)$, for some i and α . Let $prefix(b) = j$ and $suffix(b) = \gamma$. As the position set includes one from every special set of the form $r - Suffix(\beta)$, we can choose x to be in $(i - j) - Suffix(\alpha - \gamma)$. So, $x + b$ is in $i - Suffix(\alpha)$.

Note again that, if position set P and value set Q cover all the rows of C_0 , then they also cover all the rows of C_i , for all $1 \leq i \leq q-1$. That is, the columns of C_0 are the same as the columns of C_i , they are just shifted.

In Section 4 we give a specific *distance - d* partition system using a maximum matching in a bipartite graph.

In Section 5 we refine Method 2 by varying the size of the position set and symbol set, allowing many cosets to be partially covered. A dynamic programming (DP) algorithm is given that maximizes the total number of permutations by distributing the elements of the covering according to this method. It is shown in Table 14 that building PAs with partially covered cosets results in larger PAs.

3 Even Powers

Carefully choosing the position sets to complete an *incomplete distance - d* partition system we obtain a new lower bound on $M(q+1, q)$, for even powers of a prime, i.e. $q = p^{2m}$, for some m .

As $P\&E$ can create a PA with at most one more than the square root of p^m blocks, each with p^m permutations on $p^m + 1$ symbols, the construction is optimal for this technique.

Theorem 3. *For every prime p and integer $m \geq 1$, $M(p^{2m} + 1, p^{2m}) \geq (p^m + 1)p^{2m}$.*

Proof. Let $q = p^{2m}$, and let $r = p^m$. Partition the group $AGL(1, q)$ into cosets $\mathcal{C} = \{C_0, C_1, C_2, \dots, C_{q-2}\}$, as described in Section 2. Let $\Pi = (\mathcal{M}, \mathcal{P}, \mathcal{Q})$ be an *incomplete distance - d* partition system.

Each P_i has one element from each suffix set, but which element from each suffix set has not been specified.

To do this create a bipartite graph $G = (A, B, E)$, where $A = \{ (i, j) \mid 0 \leq i, j < p^m \}$ and $B = GF(q)$ are each sets of p^{2m} vertices. The node (i, j) denotes the j^{th} choice for the set P_i , which needs to be some element in $GF(q)$ whose suffix is β_j . Any element with suffix β_j will work, so a node (i, j) is connected by an edge to all such nodes. Let E be a collection of edges connecting vertices in A with vertices in B , which we now define. There is an edge connecting each vertex $(0, j)$ in A with each element in Q_j in B . Whichever edge connecting $(0, j)$ to an element of Q_j is chosen in the matching defines the position of the element with suffix β_j that is chosen in P_0 . (The position is the name of the column in C_0 .) Similarly, for $i > 0$, there is an edge connecting each vertex (i, j) in A with each element in $\{\text{shift}(i, s) \mid s \in Q_j\}$ in B . Whichever edge connecting (i, j) to an element of Q_j is chosen in the matching defines the position of the element with suffix β_j that is chosen in P_i . (The position is the name of the column in C_0 .) Thus an A-perfect matching between A and B defines each set P_i , for all i , and ensures that P_i intersected with P_j is empty (by the property of a matching). Observe that each vertex in A has r incident edges, and each vertex in B has r incident edges. It follows, as we shall see, by Hall's theorem, that there is a perfect matching in H .

It follows that the collection of sets $\mathcal{P} = \{P_0, P_1, \dots, P_{m-1}\}$ are pairwise disjoint and their union is $GF(q)$. So, $\mathcal{P}$ is a partition of $GF(q)$.

As noted earlier, for all i ($0 \leq i \leq m$), $hd(C_i) = q$ and for all i, j ($0 \leq i, j \leq m, i \neq j$), $hd(C_i, C_j) = q - 1$, so $\Pi = (\mathcal{M}, \mathcal{P}, \mathcal{Q})$ is a distance- q partition system.

So by Theorem 1, $ext(\Pi)$ is a PA on $q + 1 = p^{2m} + 1$ symbols, and $hd(ext(\Pi)) = q = p^{2m}$. Note that for all i , $|P_i| = |Q_i| = p^{2m}$, and $|C_m| = p^{2m}$. Furthermore, as shown above, $\mathcal{M}$ is well covered by $\mathcal{P}$ and $\mathcal{Q}$. So by Corollary 2, $M(p^{2m} + 1, p^{2m}) \geq |C_m| + \sum_{i=0}^{m-1} |P_i| \cdot |Q_i| = p^{2m} + p^m(p^m \cdot p^m) = (p^m + 1)p^{2m}$. $\square$

Example 1. Consider $GF(3^2)$. The elements of $GF(3^2)$, given by a generating element x , using the primitive polynomial $x^2 + x + 2$, are given in Table 2. The partition of the vertices into suffix sets is given in Table 3.

The edges in the bipartite graph for $GF(3^2)$ are given in Table 4. The nodes $(0,0)$, $(0,1)$, $(0,2)$, for the three elements in P_0 are connected to suffix 0, suffix 1, and suffix 2 nodes, respectively. The nodes $(1,0)$, $(1,1)$, $(1,2)$, for the three elements in P_1 , are connected to $shift(1, s)$ nodes, with s in suffix 0, $shift(1, t)$ nodes, with t in suffix 1, $shift(1, u)$ nodes, with u in suffix 2, respectively. Finally, the nodes $(2,0)$, $(2,1)$, $(2,2)$, for the three elements in P_2 , are connected to $shift(2, s)$ nodes, with s in suffix 0, $shift(2, t)$ nodes, with t in suffix 1, $shift(2, u)$ nodes, with u in suffix 2, respectively.

For example, we could choose the perfect matching given in Table 5.

This means that $P_0 = \{1, 4, 6\}$, $P_1 = \{0, 2, 3\}$, i.e. $shift(1, x)$, for $x \in R_1 = \{0, 3, 4\}$, $P_2 = \{5, 7, 8\}$, i.e. $shift(2, x)$, for $x \in R_2 = \{1, 2, 7\}$. As can be seen, each of the sets P_0, R_1 , and R_2

index	power of x	remainder polynomial	remainder coefficients
0	0	0	(0,0)
1	x^0	1	(0,1)
2	x^1	x	(1,0)
3	x^2	$2x + 1$	(2,1)
4	x^3	$2x + 2$	(2,2)
5	x^4	2	(0,2)
6	x^5	$2x$	(2,0)
7	x^6	$x + 2$	(1,2)
8	x^7	$x + 1$	(1,1)

Table 2: Elements of $GF(3^2)$

suffix 0	suffix 1	suffix 2
0 =(0,0)	1 =(0,1)	5 =(0,2)
2 =(1,0)	8 =(1,1)	7 =(1,2)
6 =(2,0)	3 =(2,1)	4 =(2,2)

Table 3: Partition of $GF(3^2)$ by suffixes.

node in A	connected to	node in A	connected to	node in A	connected to
(0,0)	0,2,6	(1,0)	0,1,5	(2,0)	0,4,8
(0,1)	1,3,8	(1,1)	2,7,8	(2,1)	1,6,7
(0,2)	4,5,7	(1,1)	3,4,6	(2,2)	2,3,5

Table 4: Edges in the bipartite graph for $GF(3^2)$

node in A	matched with	node in A	matched with	node in A	matched with
(0,0)	6	(1,0)	0	(2,0)	8
(0,1)	1	(1,1)	2	(2,1)	7
(0,2)	4	(1,1)	3	(2,2)	5

Table 5: Matching in the bipartite graph for $GF(3^2)$

have one element from each suffix set, and the sets P_0, P_1, P_2 are pairwise disjoint. If Q_0, Q_1, Q_2 are the set of all strings with suffix 0, 1, 2, respectively, then (as we have seen) cosets C_0, C_1, C_2 are well covered by $(P_0, Q_0), (P_1, Q_1), (P_2, Q_2)$. Together with the freebie coset, four cosets are fully covered, so we have $M(10, 9) \geq 36$. (In [15] Janiszczak and Staszewski showed $M(10, 9) \geq 49$.)

As noted above, we use a matching procedure to determine the sets in $\mathcal{P}$, the partition of positions. We use Hall's theorem on a perfect matching in a bipartite graph, with bipartite sets X and Y , which we state here for completeness. For a set W of vertices in X , let $N(W)$ denote the neighborhood of W , *i.e.*, the set of all vertices in Y adjacent to some element of W .

Theorem 4 (Hall's Theorem [21]). *Let G be a finite bipartite graph with bipartite sets X and Y . A necessary and sufficient condition for there to be a complete matching from X to Y in G is that*

$|W| \leq |N(W)|$ for every subset W of X .

Note that, as every node in both X and Y has degree d , for some fixed d , the conditions of Hall's theorem are satisfied. That is, if there were a subset W of X such that $|W| > |N(W)|$, then by the Pigeonhole Principle, since there are a total of $|W|d$ connections between W and $N(W)$, there must be a node in $N(W)$ with at least $|W|d/|N(W)| > d$ incident edges. This contradicts the fixed degree assumption. So, no subset W such that $|W| > |N(W)|$ exists.

As an example of Theorem 3, $M(65, 64) \geq 1088$, improving the previous result, $M(65, 64) \geq 576$ [7].

4 Odd Powers

Construct a bipartite graph $G = (A, B, E)$ as follows. Partition the left vertex set as $A = A_1 \cup A_2$, where

$$\begin{aligned} A_1 &= \{(i, \alpha) : 1 \leq i \leq S, \alpha \text{ is a length-}m \text{ string}\}, \\ A_2 &= \{(i, j, \beta) : S + 1 \leq i \leq U, j - \text{Suffix}(\beta) \text{ is a special set}\}. \end{aligned}$$

The right vertex set is

$$B = GF(q).$$

For $v \in A$, define

$$N(v) = \begin{cases} \text{shift}(i - 1, \text{Number}(\text{Suffix}(\alpha))), & v = (i, \alpha) \in A_1, \\ \text{shift}(i - 1, \text{Number}(j - \text{Suffix}(\beta))), & v = (i, j, \beta) \in A_2. \end{cases}$$

Finally, let

$$E = \{\{v, x\} : v \in A, x \in N(v)\}.$$

We show that a maximum matching always exists in the bipartite graph G using Theorem 4.

Suppose there were a set X with n nodes such that $|\Gamma(X)| < |X|$. There are two types of nodes in the set A . There are degree p^{m+1} nodes of the form (i, α) , called *Method 1 nodes*, corresponding to a use of Method 1, and there are degree p^m nodes of the form (i, j, α) , called *Method 2 nodes*, corresponding to a use of Method 2.

Let x_i be the number of Method 1 nodes in X corresponding to coset i . Then, $x_i p^{m+1} \leq n - 1$. So, $x_i \leq \frac{n-1}{p^{m+1}}$. Similarly, let y_i be the number of Method 2 nodes in X corresponding to coset i . Then, $y_i p^m \leq n - 1$. So, $y_i \leq \frac{n-1}{p^m}$.

Recall that $S = \left\lfloor \frac{p^{n+1}}{p+1} \right\rfloor$ and that there are S nodes for method 1 and S nodes for method 2, so $n = \sum_i x_i + \sum_i y_i \leq S(n - 1)(\frac{1}{p^{m+1}} + \frac{1}{p^m}) = S(n - 1)\frac{p+1}{p^{m+1}} \leq n - 1$. This is a contradiction. So, there is no subset X of the set of nodes A in G such that $|X| > |\Gamma(X)|$. By Theorem 4 there is a perfect A matching in the bipartite graph G .

Putting all the pieces together, as we have a maximum matching in the bipartite graph G there is a partition of the set $GF(q)$ into disjoint position sets, one for each of the $U = 2S = 2\frac{p^{m+1}}{p+1}$ cosets. As seen, this covers all of the permutations in the U cosets. So, if one divides p^{m+1} by $p+1$, one obtains the integer S with a remainder of r . $U = 2S$ is the number of cosets fully covered by the partition system using Method 1 and, separately, by Method 2. One also includes the *freebie* coset in the collection of extended sets. Thus, we have $1 + 2S$ cosets, each of size p^{2m+1} .

The remainder indicates the number of positions and values left over. If m is even, the remainder is p and, if m is odd, the remainder is 1. If m is even, the remainder p means there is one position left in each of the special sets and p special sets for values. Using Method 2 one can cover p^4 permutations in an additional coset. If m is odd, the remainder 1 means there is one special set left over for values and one from each suffix set for positions. Using Method 1 one can cover p^2 permutations in an additional coset.

This is summarized in Theorem 5.

Theorem 5. For any prime p and $m \geq 1$, where $p^{m+1} = S * (p + 1) + r$, with $0 \leq r \leq p$, $M(1 + p^{2m+1}, p^{2m+1}) \geq \Phi = (1 + 2S)p^{2m+1} + (rp)^2$.

Example 2. Consider $GF(3^3)$. The elements of $GF(3^3)$, given by a generating element x , using the primitive polynomial $x^3 + 2x^2 + 1$, are given in Table 6. The partition of $GF(3^3)$ into prefixes and suffixes is given in Table 7. The edges of the bipartite graph are given in Table 8. The matching is given in Table 9.

So, coset C_0 is fully covered by the position set $P_0 = \{0, 1, 4\}$ and the value set $Q_0 = Suffix(0)$, C_1 is fully covered by the position set $P_1 = \{2, 3, 7\}$ and the value set $Q_1 = Suffix(1)$, C_2 is fully covered by position set $P_2 = \{5, 6, 10, 11, 13, 15, 19, 22, 25\}$ and the value set $Q_2 = 0 - Suffix(2)$, C_3 is fully covered with the position set $P_3 = \{8, 9, 12, 14, 16, 18, 21, 23, 26\}$ and the value set $Q_3 = 1 - Suffix(2)$. This is shown in the Appendix. Note that the position sets are pairwise disjoint, as they need to be for $P\&E$. Note also that there are $p = 3$ positions not included and $p = 3$ values not included. The remaining set of positions is $\{17, 20, 24\}$, a set consisting of one permutation from each suffix set, and the remaining set of values is $2 - Suffix(2)$. These can be used to cover nine additional permutations, say, in coset C_4 , as shown in the Appendix. Together with coset C_5 , the so-called *freebie*, we will have extended $5 * 27 + 9 = 144$ permutations. So, $M(28, 27) \geq 144$, which improves on the previous bound of 140 obtained from five MOLS of side 28.

index	power of x	remainder	remainder coefficients	index	power of x	remainder	remainder coefficients
0	0	0	(0,0,0)	14	x^{13}	2	(0,0,2)
1	x^0	1	(0,0,1)	15	x^{14}	$2x$	(0,2,0)
2	x^1	x	(0,1,0)	16	x^{15}	$2x^2$	(2,0,0)
3	x^2	x^2	(1,0,0)	17	x^{16}	$2x^2 + 1$	(2,0,1)
4	x^3	$x^2 + 2$	(1,0,2)	18	x^{17}	$2x^2 + x + 1$	(2,1,1)
5	x^4	$x^2 + 2x + 2$	(1,2,2)	19	x^{18}	$x + 1$	(0,1,1)
6	x^5	$2x + 2$	(0,2,2)	20	x^{19}	$x^2 + x$	(1,1,0)
7	x^6	$2x^2 + 2x$	(2,2,0)	21	x^{20}	$2x^2 + 2$	(2,0,2)
8	x^7	$x^2 + 1$	(1,0,1)	22	x^{21}	$2x^2 + 2x + 1$	(2,2,1)
9	x^8	$x^2 + x + 2$	(1,1,2)	23	x^{22}	$x^2 + x + 1$	(1,1,1)
10	x^9	$2x^2 + 2x + 2$	(2,2,2)	24	x^{23}	$2x^2 + x + 2$	(2,1,2)
11	x^{10}	$x^2 + 2x + 1$	(1,2,1)	25	x^{24}	$2x + 1$	(0,2,1)
12	x^{11}	$x + 2$	(0,1,2)	26	x^{25}	$2x^2 + 2x$	(2,2,0)
13	x^{12}	$x + 2$	(1,2,0)				

Table 6: Elements of $GF(3^3)$.

prefix	suffix 0	suffix 1	suffix 2
0	0 =(0,0,0) 13 =(1,2,0) 26 =(2,1,0)	1 =(0,0,1) 11 =(1,2,1) 18 =(2,1,1)	5 =(1,2,2) 14 =(0,0,2) 24 =(2,1,2)
1	2 =(0,1,0) 3 =(1,0,0) 7 =(2,2,0)	19 =(0,1,1) 8 =(1,0,1) 22 =(2,2,1)	12 =(0,1,2) 4 =(1,0,2) 10 =(2,2,2)
2	15 =(0,2,0) 16 =(2,0,0) 20 =(1,1,0)	25 =(0,2,1) 17 =(2,0,1) 23 =(1,1,1)	6 =(0,2,2) 21 =(2,0,2) 9 =(1,1,2)

Table 7: Partition of $GF(3^3)$ by prefixes and suffixes

node in A	connects to	node in A	connects to
(1,Suffix(0))	0,2,3,7,13,15,16,20,26	(3,0,Suffix(2))	3,12,22
(1,Suffix(1))	1,8,11,17,18,19,22,23,25	(3,1,Suffix(2))	2,8,10
(1,Suffix(2))	4,5,6,9,10,12,14,21,24	(3,2,Suffix(2))	4,7,19
(2,Suffix(0))	0,1,2,6,12,14,15,19,25	(4,0,Suffix(0))	0,10,23
(2,Suffix(1))	26,7,10,16,17,18,21,22,24	(4,1,Suffix(0))	25,26,4
(2,Suffix(2))	3,4,5,8,9,11,13,20,23	(4,2,Suffix(0))	12,13,17
(3,0,Suffix(0))	0,11,24	(4,0,Suffix(1))	24,8,15
(3,1,Suffix(0))	26,1,5	(4,1,Suffix(1))	5,16,19
(3,2,Suffix(0))	13,14,18	(4,2,Suffix(1))	14,20,22
(3,0,Suffix(1))	25,9,16	(4,0,Suffix(2))	2,11,21
(3,1,Suffix(1))	6,17,20	(4,1,Suffix(2))	1,7,9
(3,2,Suffix(1))	15,21,23	(4,2,Suffix(2))	3,6,18

Table 8: Edges in the bipartite graph for $GF(3^3)$

node in A	matched with	node in A	matched with
(1,Suffix(0))	0	(3,0,Suffix(2))	22
(1,Suffix(1))	1	(3,1,Suffix(2))	10
(1,Suffix(2))	4	(3,2,Suffix(2))	19
(2,Suffix(0))	2	(4,0,Suffix(0))	23
(2,Suffix(1))	7	(4,1,Suffix(0))	26
(2,Suffix(2))	3	(4,2,Suffix(0))	12
(3,0,Suffix(0))	11	(4,0,Suffix(1))	8
(3,1,Suffix(0))	5	(4,1,Suffix(1))	16
(3,2,Suffix(0))	13	(4,2,Suffix(1))	14
(3,0,Suffix(1))	25	(4,0,Suffix(2))	21
(3,1,Suffix(1))	6	(4,1,Suffix(2))	9
(3,2,Suffix(1))	15	(4,2,Suffix(2))	18

Table 9: Matching in the bipartite graph for $GF(3^3)$

5 Dynamic Programming for Odd Powers

The method in Section 4 fully covered its cosets. In this section, we allow individual cosets to be only partially covered in order to cover more permutations overall, often resulting in higher bounds. For example, when $q = 5^3$, this technique gives a bound of $M(126, 125) \geq 1375$, which is an improvement over the bound of 1150 found by the method described in Section 4.

Throughout this section, we will illustrate various concepts using $q = 5^3$ as an example. The elements of the Galois field can be found in Table 17 in the appendix. The special sets can be found in Table 10.

$i \setminus \beta$	0	1	2	3	4
0	{0, 5, 36, 67, 98}	{1, 53, 85, 87, 110}	{22, 54, 56, 79, 94}	{17, 32, 84, 116, 118}	{23, 25, 48, 63, 115}
1	{2, 3, 12, 78, 90}	{10, 24, 51, 75, 104}	{52, 57, 60, 102, 120}	{15, 31, 37, 50, 70}	{7, 14, 18, 35, 61}
2	{47, 59, 95, 96, 105}	{6, 19, 39, 108, 124}	{20, 44, 73, 103, 117}	{4, 30, 100, 107, 111}	{21, 26, 29, 71, 89}
3	{33, 34, 43, 109, 121}	{9, 27, 83, 88, 91}	{38, 45, 49, 66, 92}	{11, 41, 55, 82, 106}	{46, 62, 68, 81, 101}
4	{16, 28, 64, 65, 74}	{69, 76, 80, 97, 123}	{8, 77, 93, 99, 112}	{40, 58, 114, 119, 122}	{13, 42, 72, 86, 113}

Table 10: Special sets of $GF(5^3)$. Rows give prefix residues from the highest 2 coefficient(s), and columns give suffixes from the lowest 1 coefficient(s).

5.1 Partial Coverage of Cosets

Let a stripe of prefix residue r for coset C_i be a set of p^m elements such that, after shifting the elements according to the properties of coset i , each of the possible suffixes of length m is represented, and all of the elements have the same prefix residue r . We can represent this using the *shift* function as $S = \{\text{shift}(i, z_{(r, \beta)}) : \beta \in \mathbb{Z}_p^m\}$, where $z_{(r, \beta)}$ is some element in $r\text{-Suffix}(\beta)$. We presently specify only the required properties of a stripe, and we delay the method of choosing disjoint elements to satisfy these properties until Section 5.3.

When performing P&E, we restrict our attention to a family of coverings determined by three integers a, b, t , with $1 \leq a, b, t \leq p$ in order to obtain a tractable search space. For any coset, the position set will be a union of a stripes, and the symbol set will be a union of b special sets with prefix residues that are spaced t apart modulo p . Let any pair (P_i, Q_i) that covers a coset C_i according to these properties be called an (a, b, t) -covering.

For some P_i that is a union of a stripes, the prefix residues are chosen consecutively modulo p . In other words, for some starting prefix residue x_0 , the stripes have prefix residues $x_0, x_0 + 1, \dots, x_0 + a - 1 \pmod{p}$. So if a coset with $q = 3^3 = 27$ uses $a = 2$ stripes, then those prefix residues might be $\{0, 1\}$. Or if we need to wrap around modulo p , for example, a coset with $q = 5^3 = 125$ using $a = 3$ stripes might use prefix residues $\{4, 0, 1\}$.

For some Q_i that is a union of b special sets, the prefix residues are chosen to be t apart modulo p . In other words, for some starting prefix residue y_0 , the special sets have prefix residues $y_0, y_0 + t, \dots, y_0 + (b - 1)t \pmod{p}$. So if a coset with $q = 7^3 = 343$ uses $b = 4$ special sets with prefix residues that are $t = 2$ apart, then those prefix residues might be $\{0, 2, 4, 6\}$, and the resulting set of symbols will be:

$$0\text{-Suffix}(1) \cup 2\text{-Suffix}(1) \cup 4\text{-Suffix}(1) \cup 6\text{-Suffix}(1)$$

If we need to, we will wrap around modulo p when adding t . For example, a coset with $q = 7^3 = 343$ might use 5 special sets with suffix 1 and with prefix residues 2 apart, then those prefix

residues might be $\{3, 5, 0, 2, 4\}$, and the symbols be:

$$3\text{-Suffix}(1) \cup 5\text{-Suffix}(1) \cup 0\text{-Suffix}(2) \cup 2\text{-Suffix}(2) \cup 4\text{-Suffix}(2)$$

In example above, notice that the suffix changes from 1 to 2 when the prefix residue wraps. We will see in Lemma 6 that the prefix residue is the key to predicting coverage. Therefore, the suffixes of the special sets are irrelevant to coverage. For ease of computation, we increment the suffix whenever the prefix residue wraps around, but this is a minor implementation detail, and any choice of suffixes would yield the same coverage.

Lemma 6. *Let P_i be a single stripe of prefix residue u for coset C_i , and let $Q_i = v\text{-Suffix}(\beta)$ be a single special set. Then (P_i, Q_i) covers exactly those permutations $\pi_{i,j} \in C_i$ for which $\text{prefix}(j) = v - u$.*

In particular, exactly p^{2m} permutations of C_i are covered.

Proof. First, consider the case for C_0 . Let π_j be the j 'th permutation of C_0 , written as $x + j$ under the field addition. This π_j is covered exactly when there is some x in P_0 such that $x + j \in v\text{-Suffix}(\beta)$. Since addition on elements of $GF(p^{2m+1})$ is performed coordinate-wise, $\text{suffix}(x + j) = \text{suffix}(x) + \text{suffix}(j)$. Since P_0 is a stripe, every possible suffix is represented by some element $x \in P_0$. So there always exists some $x \in P_0$ such that $\text{suffix}(x) + \text{suffix}(j) = \beta$. Therefore, to determine when π_j is covered, it only remains to determine when $\text{prefix}(x + j) = v$.

Coordinate-wise addition means $\text{prefix}(x + j) = \text{prefix}(x) + \text{prefix}(j)$. Thus π_j is covered exactly when $\text{prefix}(x) + \text{prefix}(j) = v$ has a solution. Since any $x \in P_0$ has prefix residue u , π_j is covered if and only if $\text{prefix}(j) = v - u$. Therefore, the $(1, 1, 1)$ -covering covers exactly those p^{2m} permutations π_j for which $\text{prefix}(j) = v - u$.

This method of counting results in the same coverage for any coset C_i because in the definition of a stripe its positions have the form $\text{shift}(i, x)$, where $\text{prefix}(x) = u$ and every possible suffix of x is represented. Since $\pi_{i,j}(\text{shift}(i, x)) = \text{shift}^{-1}(i, \text{shift}(i, x)) + j = x + j$, the preceding argument applies without change. Therefore $\pi_{i,j}$ is covered if and only if $\text{prefix}(j) = v - u$. $\square$

We have seen in Lemma 6 how the coverage of a single stripe and a single special set is determined entirely by their difference in prefix residue. We will see shortly how the same is true for any any (a, b, t) -covering. However, since a covering with multiple stripes or multiple special sets may produce the same prefix residue difference in more than one way, we must first find a means of accounting for this overlap.

We noted before that the prefix residues chosen in P_i or Q_i could be described by some starting residue of x_0 or y_0 respectively. Consider then the set of prefix residue differences:

$$\{(y_0 + yt) - (x_0 + x) \pmod{p} : x \in [0..a - 1], y \in [0..b - 1]\}$$

Since a change in either starting residue's value can only translate the various differences around the addition table of $\mathbb{Z}_p$, the starting residues do not affect the size of the set. Therefore, without loss of generality, we can assume $x_0 = y_0 = 0$ and count the number of differences with a function which only depends on a, b, t . Let

$$\omega(a, b, t) = |\{y * t - x \pmod{p} : x \in [0..a - 1], y \in [0..b - 1]\}|$$

Lemma 7. *The number of permutations covered by the (a, b, t) -covering is $p^{2m} * \omega(a, b, t)$.*

Proof. The permutations $\pi_{i,j} \in C_i$ can be partitioned into p disjoint classes of size p^{2m} based on $\text{prefix}(j)$. By Lemma 6, every permutation in the class for $\text{prefix}(r)$ is covered exactly when $r = v - u$ for some stripe with prefix residue u and some special set with prefix residue v . There are exactly $\omega(a, b, t)$ distinct prefix residue differences, so the total coverage is $p^{2m} * \omega(a, b, t)$. $\square$

We can apply this technique to every coset that we cover in $AGL(1, q)$. The skipping factor t raises the question of how to pack our special sets densely so that none go to waste. We make the simplifying assumption that each choice of triple (a, b, t) is applied to t consecutive cosets, so that we can interleave the choices in an efficient way. We will call this collection of cosets an (a, b, t) -block. Each (a, b, t) -block uses at stripes and bt special sets to cover t cosets. Since each of the t cosets covers $p^{2m} * \omega(a, b, t)$ permutations, the entire block contributes $t * p^{2m} * \omega(a, b, t)$ covered permutations.

From this construction and lemma 7, it immediately follows that a $P\&E$ system made entirely of (a, b, t) -blocks has a coverage of:

$$p^{2m} \sum_i t * \omega(a_i, b_i, t_i)$$

5.2 Dynamic Programming

The advantage of using these (a, b, t) -blocks over the technique in Section 4 is that the distribution of positions and symbols can be optimized, potentially achieving higher bounds. We optimize the coverage as stated in the previous section using a dynamic programming approach.

First, we recast the sum as a recurrence, factoring out the p^{2m} term for simplicity:

$$\theta(u, v) = \begin{cases} 0, & u = 0 \text{ or } v = 0, \\ \max_{\substack{1 \leq a, b, t \leq p \\ at \leq u, bt \leq v}} \{t \omega(a, b, t) + \theta(u - at, v - bt)\}, & u, v > 0. \end{cases}$$

Lemma 8. *The maximum coverage attainable using (a, b, t) -blocks with u available stripes and v available special sets is $p^{2m} * \theta(u, v)$.*

Proof. By strong induction on $u + v$. For the induction hypothesis assume the lemma holds for all pairs u', v' such that $u' + v' < u + v$.

Boundary case. $u = 0$ or $v = 0$. No stripes or no special sets are available to cover any permutations. The induction holds since $\theta(u, v) = 0$.

Inductive step. $u, v > 0$. Consider an optimal covering $(\mathcal{P}, \mathcal{Q})$ which uses u stripes and v special sets.

Since the covering is not empty, it must contain at least one (a, b, t) -block. By Lemma 7, this block yields $t * p^{2m} * \omega(a, b, t)$ permutations. Removing this block from $(\mathcal{P}, \mathcal{Q})$ leaves us with $u - at$ stripes and $v - bt$ special sets. Let $(\mathcal{P}^-, \mathcal{Q}^-)$ be the remaining covering. $(\mathcal{P}^-, \mathcal{Q}^-)$ must provide an optimal covering using the remaining stripes and sets; otherwise, there would be a better covering, call it $(\mathcal{P}', \mathcal{Q}')$, and adding our (a, b, t) -block to $(\mathcal{P}', \mathcal{Q}')$ would yield better coverage than $(\mathcal{P}, \mathcal{Q})$, contradicting our assumption that $(\mathcal{P}, \mathcal{Q})$ is optimal.

Since the smaller covering $(\mathcal{P}^-, \mathcal{Q}^-)$ is optimal among all coverings that use (a, b, t) -blocks and use at most $u - at$ stripes and $v - bt$ special sets, we can invoke the induction hypothesis to find

that it yields a coverage of $p^{2m} * \theta(u - at, v - bt)$. Since we know the covering of an (a, b, t) -block is $p^{2m} * t * \omega(a, b, t)$, the optimal covering $(\mathcal{P}, \mathcal{Q})$ has a coverage of $p^{2m} (t * \omega(a, b, t) + \theta(u - at, v - bt))$. By the definition of θ , this value is equal to $\theta(u, v)$. Therefore the maximum coverage attainable using (a, b, t) -blocks with u available stripes and v available special sets is $p^{2m} * \theta(u, v)$, and the inductive step is complete. $\square$

We give pseudocode for the dynamic programming algorithm in Figure 1.

```

1  Algorithm dp_split(p,m)
2
3  Input:
4    prime p and integer m > 0
5
6    B      ← p^(m+1)
7    A      ← (B+1)*(B+1) sized array, initialized to all zeros
8    choice ← (B+1)*(B+1) sized array, initialized to all null
9
10   for u in 1..B:
11     for v in 1..B:
12       for a in 1..min(p, u):
13         for b in 1..min(p, v):
14           limit ← min(p, floor(u/a), floor(v/b))
15           for t in 1..limit:
16             gain ← t * \omega(a, b, t)
17             candidate ← gain + A[u-t*a, v-t*b]
18             if candidate > A[u,v]:
19               A[u,v] ← candidate
20               choice[u,v] ← (a, b, t)
21
22   split ← empty list
23   u = B
24   v = B
25   while choice[u,v] not null:
26     a, b, t = choice[u,v]
27     append (a,b,t) to split
28     u ← u-a*t
29     v ← v-b*t
30   return A[B,B], split

```

Figure 1: Pseudocode for dynamic programming.

The correctness of the above algorithm follows from the recurrence and optimal substructure proven in Lemma 8.

We see from the size of the arrays that the space complexity is $O(p^{2m+2})$. By multiplying the ranges of the nested ‘for’ loops, we see that the complexity of filling the table is $O(p^{2m+5})$ if $\omega(a, b, t)$ is pre-computed.

block	(a, b, t)	cosets	at	bt	ω	covered
1	(2, 2, 2)	C_0-C_1	4	4	4	200
2	(2, 2, 3)	C_2-C_4	6	6	4	300
3	(2, 3, 3)	C_5-C_7	6	9	5	375
4	(3, 2, 3)	C_8-C_{10}	9	6	5	375
Σ			25	25		1250

Table 11: Output of the dynamic program for $GF(5^3)$. Adding the freebie of size $5^3 = 125$ yields a total coverage of 1375.

5.3 Constructing a $P\&E$ System

Just like in Section 2, our dynamic program determines the shape of our $P\&E$ system without assigning the elements explicitly. Now, as in Section 3 and 4, we will assign the Q symbols in a straightforward way, then use bipartite matching to achieve the more difficult assignment of positions P .

5.3.1 Symbols

We assign the special sets to the sets in $\mathcal{Q}$ in a simple way. First, we define a sequence R in which all special sets are sorted into groups by suffix in ascending order and sorted by prefix residue within each group. For example, the special sets of 3^3 would be $R_0 = 0\text{-}Suffix(0)$, $R_1 = 1\text{-}Suffix(0)$, $R_2 = 2\text{-}Suffix(0)$, $R_3 = 0\text{-}Suffix(1)$, $R_4 = 1\text{-}Suffix(1)$, $R_5 = 2\text{-}Suffix(1)$, $R_6 = 0\text{-}Suffix(2)$, $R_7 = 1\text{-}Suffix(2)$, and $R_8 = 2\text{-}Suffix(2)$.

Consider an (a, b, t) -block which we will use to cover cosets $C_s, \dots, C_{s+t-1}$. We take the next $b*t$ unused special sets in R and distribute them among the t symbol sets $Q_s, \dots, Q_{s+t-1}$ in a round robin. Suppose R_r were the next available special set. Then Q_s will be $R_r \cup R_{r+t} \cup \dots R_{r+(b-1)t}$, and Q_{s+1} will be $R_{r+1} \cup \dots R_{r+(b-1)t+1}$, and so on, so that Q_{s+t-1} will be $R_{r+t-1} \cup \dots R_{r+(b-1)t+t-1}$. From this, it is evident that bt different special sets are consumed by each (a, b, t) -block.

This distribution of special sets gives us the desired property that within each Q_s , the prefix residues of the special sets are t apart modulo p . Recall that the coverage of a coset is dependent on the prefix residues, not the suffixes, of the special sets.

Let $D = \{(a_1, b_1, t_1), \dots, (a_s, b_s, t_s)\}$ be the output of the DP algorithm. Let $S_r = \sum_{j < r} t_j$ be the partial sum of the t elements of the first r triples. Then the triple (a_r, b_r, t_r) represents the cosets $C_{S_r}, C_{S_r+1}, \dots, C_{S_r+t_r-1}$. Each such coset would need a_r stripes and b_r special sets.

5.3.2 Positions

The position set for any coset is a union of a_r stripes. By definition, we need one position for each of the p^m suffixes in each of the a_r stripes. Therefore, we create a bipartite graph where the left side of the graph has vertices of the form (i, g, β) for each P_i that requires an element of a stripe for coset C_i with prefix residue g and suffix β . On the right side of the graph, we create one vertex for each element of the Galois field. We use the edges to account for the shifting of columns in each coset. Construct a bipartite graph $G = (A, B, E)$ as follows:

$$A = \bigcup_{r=1}^s \{(S_r + \ell, g, \beta) : 0 \leq \ell < t_r, 0 \leq g < a_r, \beta \in \mathbb{Z}_p^m\},$$

block	symbol set	special sets
1	Q_0	R_0, R_2
1	Q_1	R_1, R_3
2	Q_2	R_4, R_7
2	Q_3	R_5, R_8
2	Q_4	R_6, R_9
3	Q_5	R_{10}, R_{13}, R_{16}
3	Q_6	R_{11}, R_{14}, R_{17}
3	Q_7	R_{12}, R_{15}, R_{18}
4	Q_8	R_{19}, R_{22}
4	Q_9	R_{20}, R_{23}
4	Q_{10}	R_{21}, R_{24}

Table 12: Round-robin assignment of special sets to the symbol sets.

The right vertex set is

$$B = GF(q)$$

For each $v = (i, g, \beta) \in A$, define

$$N(v) = \{shift(i, z) : z \in g-Suffix(\beta)\}$$

Finally, let $E = \{\{v, x\} : v \in A, x \in N(v)\}$

If G has a matching that uses every vertex in A , we assign the matching field element in B to each vertex (i, g, β) . For fixed i and g , the p^m matched elements form the stripe of prefix residue g for coset C_i required by P_i . Since a matching uses each vertex of B at most once, the resulting position sets are disjoint.

Finding a matching in this graph is not trivial. There may not be a matching in all cases, but a matching has been found in all cases tried. A small set of results are given in Table 14.

6 Extension by Two Symbols

Theorem 9. *Let $q = p^k$ where p is a prime and $k \geq 1$ is an integer. Then $M(q+2, q) \geq q(q-1)$.*

Proof. Consider C_0 and the $q-2$ cosets of the subgroup C_0 of $AGL(1, q)$, say $C_1, C_2, \dots, C_{q-2}$, where $C_i = \{(i+1)x + a \mid a \in GF(q)\}$. As we know, $hd(C_i) = q$ and $hd(C_i, C_j) = q-1$.

Create the blocks of permutations on $q+2$ symbols, $K_0, K_1, \dots, K_{q-2}$, for each i , where K_i is obtained from C_i by modifying each permutation π in C_i as follows:

- (a) adding the pair of symbols $(i, i+1)$ to the pair of positions $(q, q+1)$,
- (b) replacing the original pair of symbols $(i, i+1)$ which appear somewhere in π (not necessarily consecutively) by the pair of symbols $(q, q+1)$, respectively.

See Table 15 for an example of the described transformation from C_0 to K_0 . Note that inside K_i the Hamming distance is q , as there are no agreements between permutations in C_i , and there are now two agreements between any pair of permutations in K_i , namely at the last two positions.

The Hamming distance between any two permutations in K_i and K_j ($i \neq j$) is at least q , as we shall see. They do not agree in the last two positions, as those in K_i end with $(i, i+1)$ and those in K_j end with $(j, j+1)$, with $i \neq j$. Furthermore, the corresponding permutations in C_i and C_j have one agreement, say at some position R . Replacing the pair $(i, i+1)$ by the pair of symbols

position set	stripes selected by the matching
P_0	$g = 0 : \{0, 1, 17, 23, 54\}$ $g = 1 : \{2, 7, 10, 15, 52\}$
P_1	$g = 0 : \{4, 16, 55, 84, 114\}$ $g = 1 : \{6, 9, 11, 14, 119\}$
P_2	$g = 0 : \{3, 20, 30, 46, 83\}$ $g = 1 : \{5, 8, 35, 50, 76\}$
P_3	$g = 0 : \{33, 45, 51, 82, 113\}$ $g = 1 : \{12, 32, 48, 49, 75\}$
P_4	$g = 0 : \{18, 63, 81, 111, 112\}$ $g = 1 : \{27, 31, 47, 53, 74\}$
P_5	$g = 0 : \{43, 62, 79, 89, 120\}$ $g = 1 : \{26, 56, 70, 73, 97\}$
P_6	$g = 0 : \{61, 78, 88, 104, 109\}$ $g = 1 : \{25, 29, 69, 72, 96\}$
P_7	$g = 0 : \{41, 60, 77, 87, 103\}$ $g = 1 : \{24, 68, 71, 95, 124\}$
P_8	$g = 0 : \{40, 59, 86, 102, 108\}$ $g = 1 : \{42, 67, 94, 118, 123\}$ $g = 2 : \{13, 36, 39, 92, 100\}$
P_9	$g = 0 : \{58, 85, 101, 106, 107\}$ $g = 1 : \{28, 66, 93, 117, 122\}$ $g = 2 : \{38, 64, 80, 91, 99\}$
P_{10}	$g = 0 : \{22, 44, 57, 105, 115\}$ $g = 1 : \{21, 65, 110, 116, 121\}$ $g = 2 : \{19, 34, 37, 90, 98\}$

Table 13: Position stripes selected by the bipartite matching.

	2	3	5	7	11	13
3	28*	144*	1375*	6076*	47190*	99710*
5	192*	3483*	159375*	1990429*	61375072*	215407062*
7	1472*	90396*	19609375*	678717081*	(running)	(oom)
9	11264*	2407887*	2443359375*	(running)	(oom)	
11	89088*	64717704*	305224609375*	(oom)		
13	704512*	1744720803*	(oom)			
15	5619712*	47083546836*				
17	44826624*	1271040530967*				
19	358350848*	(oom)				
21	2864709632*					
23	22913482752*					
25	183274307584*					
27	1466127351808*					
29	(oom)					

Table 14: Results using DP. Row k column p shows $M(p^k + 1, p^k)$. Numbers with an asterisk have been computed by DP, but the PA has not been constructed.

$(q, q + 1)$ in a permutation σ in C_i and the pair $(j, j + 1)$ by $(q, q + 1)$ in a permutation τ in C_j , we shall see makes at most one new agreement. Suppose it makes two new agreements, say at positions U and V , with $U \neq R$ and $V \neq R$, in both σ and τ . If U or V is equal to R , as our assumption is $\sigma(R) = \tau(R)$, the replacement creates at most one new agreement. Since C_i and C_j have one agreement, namely at position R , new agreements in K_i and K_j must be created by the exchange

of symbols with q and $q + 1$. So, if U and V are positions of two new agreements between σ and τ , then both σ (respectively, τ) has the symbols $\{i, i + 1\}$ (resp., $\{j, j + 1\}$) at positions $\{U, V\}$. If i (resp., j) appears before $i + 1$ (resp., $j + 1$) they are in *forward* order; otherwise, they are in *backward* order. If $i, i + 1$ are in a different order than $j, j + 1$, then the replacements by $q, q + 1$ do not make any new agreement, as q is at position U (resp., V) in one and $q + 1$ is at position U (resp., V) in the other. So, if we suppose they make two new agreements, they must have the same order. We consider the case where they are both in forward order. The case when they are both in backward order is similar. It follows from their definitions that $(i + 1)V + a - ((i + 1)U + a) = (V - U)(i + 1) = 1$ is the same as $(j + 1)V + b - ((j + 1)U + b) = (V - U)(j + 1) = 1$. That is, the difference between $\sigma(U)$ and $\sigma(V)$ is 1 and the difference between $\tau(U)$ and $\tau(V)$ is also 1. But, if $(V - U)(i + 1) = (V - U)(j + 1)$, it follows that $i = j$. This contradicts our initial assumption. So, there is at most one new agreement. Together with the existing agreement at position R , there are at most two agreements. The Hamming distance between K_i and K_j , which consists of permutations on $q + 2$ symbols, is therefore at least q . $\square$

0	1	2	3	4	5	6	7	8	2	3	4	5	6	0	1
1	5	4	2	0	6	3	8	5	4	2	7	6	3	0	1
2	4	6	5	3	0	1	2	4	6	5	3	7	8	0	1
3	2	5	1	6	4	0	3	2	5	8	6	4	7	0	1
4	0	3	6	2	1	5	4	7	3	6	2	8	5	0	1
5	6	0	4	1	3	2	5	6	7	4	8	3	2	0	1
6	3	1	0	5	2	4	6	3	8	7	5	2	4	0	1

Table 15: Coset C_0 and the resulting set K_0 for $GF(7)$.

For example, we have $M(45, 43) \geq 1806$, where the previous lower bound was 1722, obtained by sequential $P\&E$ [7]. In fact, Theorem 9 improves on all of the results given in Table 4, page 13, of [7].

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References

- [1] A. Abdollahi, J. Bagherian, F. Jafari, M. Khatami, F. Parvaresh, and R. Sobhani. Upper bounds on the size of permutation codes with a hamming distance of five. *Cryptography and Communications*, 17:1075–1091, 2025.
- [2] C. Amarra, D. V. A. Briones, and M. J. C. Loquias. Multiple contractions of permutation arrays. *Journal of Algebraic Combinatorics*, 58(3):695–715, 2023.

- [3] J. Barta and R. Montemanni. Permutation codes, hamming graphs and turán graphs. In *International Conference on Applied Physics, System Science and Computers*, pages 111–118, 2019.
- [4] J. Barta, R. Montemanni, and D. H. Smith. A branch and bound approach to permutation codes. In *2nd International Conference on Information and Communication Technology*, pages 187–192, 2014.
- [5] S. Bereg and P. J. Dukes. A lower bound on permutation codes of distance $n - 1$. *Des. Codes Cryptogr.*, 88(1):63–72, 2020.
- [6] S. Bereg, L. G. Mojica, L. Morales, and H. Sudborough. Kronecker product and tiling of permutation arrays for hamming distances. In *the 2017 IEEE International Symposium on Information Theory (ISIT)*, pages 2198–2202, 2017.
- [7] S. Bereg, L. G. Mojica, L. Morales, and I. H. Sudborough. Constructing permutation arrays using partition and extension. *Des. Codes Cryptogr.*, 88(2):311–339, 2020.
- [8] S. Bereg, L. Morales, and I. H. Sudborough. Extending permutation arrays: improving MOLS bounds. *Designs, Codes and Cryptography*, 83(3):661–683, 2017.
- [9] W. Chu, C. J. Colbourn, and P. Dukes. Constructions for permutation codes in powerline communications. *Designs, Codes and Cryptography*, 2004.
- [10] C. Colbourn, T. Kløve, and A. C. Ling. Permutation arrays for powerline communication and mutually orthogonal latin squares. *IEEE Transactions on Information Theory*, 50(6):1289–1291, 2004.
- [11] M. Deza and S. A. Vanstone. Bounds for permutation arrays. *Journal of Statistical Planning and Inference*, 2(2):197–209, 1978.
- [12] P. Dukes and Sawchuck. Bounds on permutation codes of distance four. *Journal of Algebraic Combinatorics*, 31:143–158, 2010.
- [13] S. Huczynska. Powerline communication and the 36 officers problem. *Phil. Trans. R. Soc. A.*, 364(1849):3199–3214, 2006.
- [14] I. Janiszczak, W. Lempken, P. R. J. Östergård, and R. Staszewski. Permutation codes invariant under isometries. *Des. Codes Cryptography*, 75(3):497–507, 2015.
- [15] I. Janiszczak and R. Staszewski. Improved bound for permutation arrays of length 10. Technical report, Institute for Experimental Mathematics, University Duisburg-Essen, 2008.
- [16] I. Janiszczak and R. Staszewski. Isometry invariant permutation codes and mutually orthogonal latin squares. *Journal of Combinatorial Designs*, 27(9):541–551, 2019.
- [17] N. Pavlidou, A. H. Vinck, J. Yazdani, and B. Honary. Power line communications: state of the art and future trends. *IEEE Communications magazine*, 41(4):34–40, 2003.
- [18] J. Quistorff. A new nonexistence result for sharply multiply transitive permutation sets. *Discret. Math.*, 288(1-3):185–186, 2004.

- [19] D. H. Smith and R. Montemanni. A new table of permutation codes. *Des. Codes Cryptogr.*, 63(2):241–253, 2012.
- [20] D. H. Smith, R. Montemanni, and S. Perkins. The use of an exact algorithm within a tabu search maximum clique algorithm. *Algorithms*, 13(10):253, 2020.
- [21] J. H. van Lindt and R. M. Wilson. *A Course in Combinatorics*. Cambridge University Press, 1992.
- [22] A. H. Vinck. Coded modulation for power line communications. *A. E. Ü. Int. J. Electron. Commun.*, 54(1):45–49, 2000.

q	p^k	Φ	q	p^k	Φ
8	2^3	28	343	7^3	4,508
27	3^3	144	1331	11^3	28,072
32	2^5	176	2197	13^3	55,094
125	5^3	1,150	3125	5^5	128,750
128	2^7	1,412	4913	17^3	162,418
243	3^5	3,240	6859	19^3	254,144

Table 16: Some Results of Theorem 5.

A Appendix

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26
1 14 19 8 **3** 13 15 22 4 20 7 5 2 11 0 25 17 21 24 12 23 16 10 9 26 6 18
2 19 15 20 9 4 14 16 23 5 21 8 6 3 12 0 26 18 22 25 13 24 17 11 10 1 7
3 8 20 16 21 10 5 15 17 24 6 22 9 7 4 13 0 1 19 23 26 14 25 18 12 11 2
4 **3** 9 21 17 22 11 6 16 18 25 7 23 10 8 5 14 0 2 20 24 1 15 26 19 13 12
5 **13** 4 10 22 18 23 12 7 17 19 26 8 24 11 9 6 15 0 3 21 25 2 16 1 20 14
6 **15** 14 5 11 23 19 24 13 8 18 20 1 9 25 12 10 7 16 0 4 22 26 3 17 2 21
7 22 16 15 6 12 24 20 25 14 9 19 21 2 10 26 13 11 8 17 0 5 23 1 4 18 3
8 4 23 17 **16** 7 13 25 21 26 15 10 20 22 3 11 1 14 12 9 18 0 6 24 2 5 19
9 **20** 5 24 18 17 8 14 26 22 1 16 11 21 23 4 12 2 15 13 10 19 0 7 25 3 6
10 **7** 21 6 25 19 18 9 15 1 23 2 17 12 22 24 5 13 3 16 14 11 20 0 8 26 4
11 5 8 22 **7** 26 20 19 10 16 2 24 3 18 13 23 25 6 14 4 17 15 12 21 0 9 1
12 **2** 6 9 23 8 1 21 20 11 17 3 25 4 19 14 24 26 7 15 5 18 16 13 22 0 10
13 11 3 7 10 24 9 2 22 21 12 18 4 26 5 20 15 25 1 8 16 6 19 17 14 23 0
14 **0** 12 4 8 11 25 10 3 23 22 13 19 5 1 6 21 16 26 2 9 17 7 20 18 15 24
15 25 0 13 5 9 12 26 11 4 24 23 14 20 6 2 7 22 17 1 3 10 18 8 21 19 16
16 17 26 0 14 6 10 13 1 12 5 25 24 15 21 7 3 8 23 18 2 4 11 19 9 22 20
17 21 18 1 **0** 15 7 11 14 2 13 6 26 25 16 22 8 4 9 24 19 3 5 12 20 10 23
18 24 22 19 **2** 0 16 8 12 15 3 14 7 1 26 17 23 9 5 10 25 20 4 6 13 21 11
19 12 25 23 **20** 3 0 17 9 13 16 4 15 8 2 1 18 24 10 6 11 26 21 5 7 14 22
20 23 13 26 24 21 4 0 18 10 14 17 5 16 9 3 2 19 25 11 7 12 1 22 6 8 15
21 **16** 24 14 1 25 22 5 0 19 11 15 18 6 17 10 4 3 20 26 12 8 13 2 23 7 9
22 10 17 25 **15** 2 26 23 6 0 20 12 16 19 7 18 11 5 4 21 1 13 9 14 3 24 8
23 9 11 18 **26** 16 3 1 24 7 0 21 13 17 20 8 19 12 6 5 22 2 14 10 15 4 25
24 **26** 10 12 19 1 17 4 2 25 8 0 22 14 18 21 9 20 13 7 6 23 3 15 11 16 5

All permutations
of Coset C_0
are covered.

25 6 1 11 **13** 20 2 18 5 3 26 9 0 23 15 19 22 10 21 14 8 7 24 4 16 12 17
26 18 7 2 12 14 21 3 19 6 4 1 10 0 24 16 20 23 11 22 15 9 8 25 5 17 13

0 2 3 4 5 6 7 **8** 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 1
 1 19 **8** 3 13 15 22 4 20 7 5 2 11 0 25 17 21 24 12 23 16 10 9 26 6 18 14
 2 15 20 9 4 14 16 **23** 5 21 8 6 3 12 0 26 18 22 25 13 24 17 11 10 1 7 19
 3 20 16 21 10 5 15 **17** 24 6 22 9 7 4 13 0 1 19 23 26 14 25 18 12 11 2 8
 4 9 21 **17** 22 11 6 16 18 25 7 23 10 8 5 14 0 2 20 24 1 15 26 19 13 12 3
 5 4 10 **22** 18 23 12 7 17 19 26 8 24 11 9 6 15 0 3 21 25 2 16 1 20 14 13
 6 14 5 **11** 23 19 24 13 8 18 20 1 9 25 12 10 7 16 0 4 22 26 3 17 2 21 15
 7 16 15 6 12 24 20 **25** 14 9 19 21 2 10 26 13 11 8 17 0 5 23 1 4 18 3 22
 8 23 **17** 16 7 13 25 21 26 15 10 20 22 3 11 1 14 12 9 18 0 6 24 2 5 19 4
 9 5 24 **18** 17 8 14 26 22 1 16 11 21 23 4 12 2 15 13 10 19 0 7 25 3 6 20
 10 21 6 **25** 19 18 9 15 1 23 2 17 12 22 24 5 13 3 16 14 11 20 0 8 26 4 7
 11 8 **22** 7 26 20 19 10 16 2 24 3 18 13 23 25 6 14 4 17 15 12 21 0 9 1 5
 12 6 9 **23** 8 1 21 20 11 17 3 25 4 19 14 24 26 7 15 5 18 16 13 22 0 10 2
 13 3 7 10 24 9 2 **22** 21 12 18 4 26 5 20 15 25 1 8 16 6 19 17 14 23 0 11
 14 12 4 **8** 11 25 10 3 23 22 13 19 5 1 6 21 16 26 2 9 17 7 20 18 15 24 0
 15 0 13 5 9 12 26 **11** 4 24 23 14 20 6 2 7 22 17 1 3 10 18 8 21 19 16 25
 16 26 0 14 6 10 13 **1** 12 5 25 24 15 21 7 3 8 23 18 2 4 11 19 9 22 20 17
 17 18 **1** 0 15 7 11 14 2 13 6 26 25 16 22 8 4 9 24 19 3 5 12 20 10 23 21
 18 22 **19** 2 0 16 8 12 15 3 14 7 1 26 17 23 9 5 10 25 20 4 6 13 21 11 24
 19 25 **23** 20 3 0 17 9 13 16 4 15 8 2 1 18 24 10 6 11 26 21 5 7 14 22 12
 20 13 26 24 21 4 0 **18** 10 14 17 5 16 9 3 2 19 25 11 7 12 1 22 6 8 15 23
 21 24 14 **1** 25 22 5 0 19 11 15 18 6 17 10 4 3 20 26 12 8 13 2 23 7 9 16
 22 17 **25** 15 2 26 23 6 0 20 12 16 19 7 18 11 5 4 21 1 13 9 14 3 24 8 10
 23 11 **18** 26 16 3 1 24 7 0 21 13 17 20 8 19 12 6 5 22 2 14 10 15 4 25 9
 24 10 12 **19** 1 17 4 2 25 8 0 22 14 18 21 9 20 13 7 6 23 3 15 11 16 5 26
 25 1 **11** 13 20 2 18 5 3 26 9 0 23 15 19 22 10 21 14 8 7 24 4 16 12 17 6
 26 7 2 12 14 21 3 **19** 6 4 1 10 0 24 16 20 23 11 22 15 9 8 25 5 17 13 18

All permutations
 of coset C_1
 are covered.

0 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 **24** 25 26 1 2
 1 8 3 13 15 22 4 20 7 5 2 11 0 25 17 21 24 12 23 16 10 9 26 6 18 **14** 19
 2 20 9 4 14 16 23 5 21 8 6 3 12 0 26 18 22 25 13 **24** 17 11 10 1 7 19 15
 3 16 21 10 5 15 17 24 6 22 9 7 4 13 0 1 19 23 26 **14** 25 18 12 11 2 8 20
 4 21 17 22 11 6 16 18 25 7 23 10 8 **5** 14 0 2 20 24 1 15 26 19 13 12 3 9
 5 10 22 18 23 12 7 17 19 26 8 **24** 11 9 6 15 0 3 21 25 2 16 1 20 14 13 4
 6 5 11 23 19 **24** 13 8 18 20 1 9 25 12 10 7 16 0 4 22 26 3 17 2 21 15 14
 7 15 6 12 24 20 25 14 9 19 21 2 10 26 13 11 8 17 0 **5** 23 1 4 18 3 22 16
 8 17 16 7 13 25 21 26 15 10 20 22 3 11 1 **14** 12 9 18 0 6 24 2 5 19 4 23
 9 24 18 17 8 **14** 26 22 1 16 11 21 23 4 12 2 15 13 10 19 0 7 25 3 6 20 5
 10 6 25 19 18 9 15 1 23 2 17 12 22 **24** 5 13 3 16 14 11 20 0 8 26 4 7 21
 11 22 7 26 20 19 10 16 2 24 3 18 13 23 25 6 14 4 17 15 12 21 0 9 1 **5** 8
 12 9 23 8 1 21 20 11 17 3 25 4 19 **14** 24 26 7 15 5 18 16 13 22 0 10 2 6
 13 7 10 24 9 2 22 21 12 18 4 26 5 20 15 25 1 8 16 6 19 17 **14** 23 0 11 3
 14 4 8 11 25 10 3 23 22 13 19 **5** 1 6 21 16 26 2 9 17 7 20 18 15 24 0 12
 15 13 5 9 12 26 11 4 24 23 **14** 20 6 2 7 22 17 1 3 10 18 8 21 19 16 25 0
 16 0 14 6 10 13 1 12 5 25 **24** 15 21 7 3 8 23 18 2 4 11 19 9 22 20 17 26
 17 1 0 15 7 11 **14** 2 13 6 26 25 16 22 8 4 9 24 19 3 5 12 20 10 23 21 18
 18 19 2 0 16 8 12 15 3 14 7 1 26 17 23 9 5 10 25 20 4 6 13 21 11 **24** 22
 19 23 20 3 0 17 9 13 16 4 15 8 2 1 18 **24** 10 6 11 26 21 5 7 14 22 12 25
 20 26 24 21 4 0 18 10 14 17 **5** 16 9 3 2 19 25 11 7 12 1 22 6 8 15 23 13
 21 14 1 25 22 **5** 0 19 11 15 18 6 17 10 4 3 20 26 12 8 13 2 23 7 9 16 24
 22 25 15 2 26 23 6 0 20 12 16 19 7 18 11 **5** 4 21 1 13 9 14 3 24 8 10 17
 23 18 26 16 3 1 **24** 7 0 21 13 17 20 8 19 12 6 5 22 2 14 10 15 4 25 9 11
 24 12 19 1 17 4 2 25 8 0 22 **14** 18 21 9 20 13 7 6 23 3 15 11 16 5 26 10
 25 11 13 20 2 18 **5** 3 26 9 0 23 15 19 22 10 21 14 8 7 24 4 16 12 17 6 1
 26 2 12 14 21 3 19 6 4 1 10 0 24 16 20 23 11 22 15 9 8 25 **5** 17 13 18 7

All permutations
 of coset C_2
 are covered.

0 4 5 6 7 8 9 10 11 **12** 13 14 15 16 17 18 19 20 21 22 23 24 25 26 1 2 3
 1 3 13 15 22 4 20 7 5 2 11 0 25 17 21 24 **12** 23 16 10 9 26 6 18 14 19 8
 2 9 4 14 16 23 5 21 8 6 3 12 0 26 18 22 25 13 24 17 11 **10** 1 7 19 15 20
 3 21 10 5 15 17 24 6 22 9 7 4 13 0 1 19 23 26 14 25 18 **12** 11 2 8 20 16
 4 17 22 11 6 16 18 25 7 23 10 8 5 14 0 2 20 24 1 15 26 19 13 **12** 3 9 21
 5 22 18 23 12 7 17 19 26 8 24 11 9 6 15 0 3 21 25 2 16 1 20 14 13 4 **10**
 6 11 23 19 24 13 8 18 20 1 9 25 **12** 10 7 16 0 4 22 26 3 17 2 21 15 14 5
 7 6 12 24 20 25 14 9 19 21 2 10 26 13 11 8 17 0 5 23 1 **4** 18 3 22 16 15
 8 16 7 13 25 21 26 15 10 20 22 3 11 1 14 12 9 18 0 6 24 2 5 19 **4** 23 17
 9 18 17 8 14 26 22 1 16 11 21 23 **4** 12 2 15 13 10 19 0 7 25 3 6 20 5 24
 10 25 19 18 9 15 1 23 2 17 12 22 24 5 13 3 16 14 11 20 0 8 26 **4** 7 21 6
 11 7 26 20 19 10 16 2 24 3 18 13 23 25 6 14 **4** 17 15 12 21 0 9 1 5 8 22
 12 23 8 1 21 20 11 17 3 25 4 19 14 24 26 7 15 5 18 16 13 22 0 **10** 2 6 9
 13 10 24 9 2 22 21 12 18 **4** 26 5 20 15 25 1 8 16 6 19 17 14 23 0 11 3 7
 14 8 11 25 10 3 23 22 13 19 5 1 6 21 16 26 2 9 17 7 20 18 15 24 0 12 **4**
 15 5 9 12 26 11 4 24 23 14 20 6 2 7 22 17 1 3 **10** 18 8 21 19 16 25 0 13
 16 14 6 10 13 1 12 5 25 24 15 21 7 3 8 23 18 2 **4** 11 19 9 22 20 17 26 0
 17 0 15 7 11 14 2 13 6 26 25 16 22 8 **4** 9 24 19 3 5 12 20 10 23 21 18 1
 18 2 0 16 8 12 15 3 14 7 1 26 17 23 9 5 **10** 25 20 4 6 13 21 11 24 22 19
 19 20 3 0 17 9 13 16 **4** 15 8 2 1 18 24 10 6 11 26 21 5 7 14 22 12 25 23
 20 24 21 4 0 18 10 14 17 5 16 9 3 2 19 25 11 7 **12** 1 22 6 8 15 23 13 26
 21 1 25 22 5 0 19 11 15 18 6 17 **10** 4 3 20 26 12 8 13 2 23 7 9 16 24 14
 22 15 2 26 23 6 0 20 **12** 16 19 7 18 11 5 4 21 1 13 9 14 3 24 8 10 17 25
 23 26 16 3 1 24 7 0 21 13 17 20 8 19 **12** 6 5 22 2 14 10 15 4 25 9 11 18
 24 19 1 17 4 2 25 8 0 22 14 18 21 9 20 13 7 6 23 3 15 11 16 5 26 10 **12**
 25 13 20 2 18 5 3 26 9 0 23 15 19 22 **10** 21 14 8 7 24 4 16 12 17 6 1 11
 26 12 14 21 3 19 6 4 1 **10** 0 24 16 20 23 11 22 15 9 8 25 5 17 13 18 7 2

All permutations
 of coset C_3
 are covered.

0 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 **21** 22 23 24 25 26 1 2 3 4
 1 13 15 22 4 20 7 5 2 11 0 25 17 21 24 12 23 16 10 9 26 6 18 14 19 8 3
 2 4 14 16 23 5 21 8 6 3 12 0 26 18 22 25 13 24 17 11 10 1 7 19 15 20 9
 3 10 5 15 17 24 6 22 9 7 4 13 0 1 19 23 26 14 25 18 12 11 2 8 20 16 21
 4 22 11 6 16 18 25 7 23 10 8 5 14 0 2 20 24 1 15 26 19 13 12 3 **9** 21 17
 5 18 23 12 7 17 19 26 8 24 11 9 6 15 0 3 21 25 2 16 1 20 14 13 4 10 22
 6 23 19 24 13 8 18 20 1 9 25 12 10 7 16 0 4 22 26 3 17 2 21 15 14 5 11
 7 12 24 20 25 14 9 19 21 2 10 26 13 11 8 17 0 5 23 1 4 18 3 22 16 15 6
 8 7 13 25 21 26 15 10 20 22 3 11 1 14 12 9 18 0 6 24 2 5 19 4 23 17 16
 9 17 8 14 26 22 1 16 11 21 23 4 12 2 15 13 10 19 0 7 25 3 6 20 5 24 18
 10 19 18 9 15 1 23 2 17 12 22 24 5 13 3 16 14 11 20 0 8 26 4 7 **21** 6 25
 11 26 20 19 10 16 2 24 3 18 13 23 25 6 14 4 17 15 12 21 0 9 1 5 8 22 7
 12 8 1 21 20 11 17 3 25 4 19 14 24 26 7 15 5 18 16 13 22 0 10 2 **6** 9 23
 13 24 9 2 22 21 12 18 4 26 5 20 15 25 1 8 16 **6** 19 17 14 23 0 11 3 7 10
 14 11 25 10 3 23 22 13 19 5 1 6 21 16 26 2 9 17 7 20 18 15 24 0 12 4 8
 15 9 12 26 11 4 24 23 14 20 6 2 7 22 17 1 3 10 18 8 **21** 19 16 25 0 13 5
 16 6 10 13 1 12 5 25 24 15 21 7 3 8 23 18 2 4 11 19 **9** 22 20 17 26 0 14
 17 15 7 11 14 2 13 6 26 25 16 22 8 4 9 24 19 3 5 12 20 10 23 21 18 1 0
 18 0 16 8 12 15 3 14 7 1 26 17 23 9 5 10 25 20 4 6 13 21 11 24 22 19 2
 19 3 0 17 9 13 16 4 15 8 2 1 18 24 10 6 11 26 21 5 7 14 22 12 25 23 20
 20 21 4 0 18 10 14 17 5 16 9 3 2 19 25 11 7 12 1 22 **6** 8 15 23 13 26 24
 21 25 22 5 0 19 11 15 18 6 17 10 4 3 20 26 12 8 13 2 23 7 9 16 24 14 1
 22 2 26 23 6 0 20 12 16 19 7 18 11 5 4 21 1 13 9 14 3 24 8 10 17 25 15
 23 16 3 1 24 7 0 21 13 17 20 8 19 12 6 5 22 2 14 10 15 4 25 9 11 18 26
 24 1 17 4 2 25 8 0 22 14 18 21 9 20 13 7 6 23 3 15 11 16 5 26 10 12 19
 25 20 2 18 5 3 26 9 0 23 15 19 22 10 21 14 8 7 24 4 16 12 17 6 1 11 13
 26 14 21 3 19 6 4 1 10 0 24 16 20 23 11 22 15 **9** 8 25 5 17 13 18 7 2 12

Nine permutations
 of coset C_4
 are covered.

index	power of t	remainder	remainder coefficients	index	power of t	remainder	remainder coefficients	index	power of t	remainder	remainder coefficients
0	0	0	(0, 0, 0)	42	t^{41}	$4x + 4$	(0, 4, 4)	84	t^{83}	$2x^2 + 3x + 3$	(2, 3, 3)
1	t^0	1	(0, 0, 1)	43	t^{42}	$4x^2 + 4x$	(4, 4, 0)	85	t^{84}	$3x^2 + 2x + 1$	(3, 2, 1)
2	t^1	x	(0, 1, 0)	44	t^{43}	$4x^2 + 3x + 2$	(4, 3, 2)	86	t^{85}	$2x^2 + 2x + 4$	(2, 2, 4)
3	t^2	x^2	(1, 0, 0)	45	t^{44}	$3x^2 + 2$	(3, 0, 2)	87	t^{86}	$2x^2 + 3x + 1$	(2, 3, 1)
4	t^3	$2x + 3$	(0, 2, 3)	46	t^{45}	$3x + 4$	(0, 3, 4)	88	t^{87}	$3x^2 + 1$	(3, 0, 1)
5	t^4	$2x^2 + 3x$	(2, 3, 0)	47	t^{46}	$3x^2 + 4x$	(3, 4, 0)	89	t^{88}	$2x + 4$	(0, 2, 4)
6	t^5	$3x^2 + 4x + 1$	(3, 4, 1)	48	t^{47}	$4x^2 + x + 4$	(4, 1, 4)	90	t^{89}	$2x^2 + 4x$	(2, 4, 0)
7	t^6	$4x^2 + 2x + 4$	(4, 2, 4)	49	t^{48}	$x^2 + 2x + 2$	(1, 2, 2)	91	t^{90}	$4x^2 + 4x + 1$	(4, 4, 1)
8	t^7	$2x^2 + 2x + 2$	(2, 2, 2)	50	t^{49}	$2x^2 + 4x + 3$	(2, 4, 3)	92	t^{91}	$4x^2 + 4x + 2$	(4, 4, 2)
9	t^8	$2x^2 + x + 1$	(2, 1, 1)	51	t^{50}	$4x^2 + 2x + 1$	(4, 2, 1)	93	t^{92}	$4x^2 + 2$	(4, 0, 2)
10	t^9	$x^2 + 1$	(1, 0, 1)	52	t^{51}	$2x^2 + 4x + 2$	(2, 4, 2)	94	t^{93}	2	(0, 0, 2)
11	t^{10}	$3x + 3$	(0, 3, 3)	53	t^{52}	$4x^2 + x + 1$	(4, 1, 1)	95	t^{94}	$2x$	(0, 2, 0)
12	t^{11}	$3x^2 + 3x$	(3, 3, 0)	54	t^{53}	$x^2 + 4x + 2$	(1, 4, 2)	96	t^{95}	$2x^2$	(2, 0, 0)
13	t^{12}	$3x^2 + x + 4$	(3, 1, 4)	55	t^{54}	$4x^2 + 4x + 3$	(4, 4, 3)	97	t^{96}	$4x + 1$	(0, 4, 1)
14	t^{13}	$x^2 + 4$	(1, 0, 4)	56	t^{55}	$4x^2 + x + 2$	(4, 1, 2)	98	t^{97}	$4x^2 + x$	(4, 1, 0)
15	t^{14}	$x + 3$	(0, 1, 3)	57	t^{56}	$x^2 + 2$	(1, 0, 2)	99	t^{98}	$x^2 + 3x + 2$	(1, 3, 2)
16	t^{15}	$x^2 + 3x$	(1, 3, 0)	58	t^{57}	$4x + 3$	(0, 4, 3)	100	t^{99}	$3x^2 + 4x + 3$	(3, 4, 3)
17	t^{16}	$3x^2 + 2x + 3$	(3, 2, 3)	59	t^{58}	$4x^2 + 3x$	(4, 3, 0)	101	t^{100}	$4x^2 + 4x + 4$	(4, 4, 4)
18	t^{17}	$2x^2 + 4x + 4$	(2, 4, 4)	60	t^{59}	$3x^2 + 3x + 2$	(3, 3, 2)	102	t^{101}	$4x^2 + 2x + 2$	(4, 2, 2)
19	t^{18}	$4x^2 + 3x + 1$	(4, 3, 1)	61	t^{60}	$3x^2 + 3x + 4$	(3, 3, 4)	103	t^{102}	$2x^2 + 2$	(2, 0, 2)
20	t^{19}	$3x^2 + 4x + 2$	(3, 4, 2)	62	t^{61}	$3x^2 + 4$	(3, 0, 4)	104	t^{103}	$x + 1$	(0, 1, 1)
21	t^{20}	$4x^2 + 3x + 4$	(4, 3, 4)	63	t^{62}	4	(0, 0, 4)	105	t^{104}	$x^2 + x$	(1, 1, 0)
22	t^{21}	$3x^2 + 2x + 2$	(3, 2, 2)	64	t^{63}	$4x$	(0, 4, 0)	106	t^{105}	$x^2 + 2x + 3$	(1, 2, 3)
23	t^{22}	$2x^2 + 3x + 4$	(2, 3, 4)	65	t^{64}	$4x^2$	(4, 0, 0)	107	t^{106}	$2x^2 + 3$	(2, 0, 3)
24	t^{23}	$3x^2 + 3x + 1$	(3, 3, 1)	66	t^{65}	$3x + 2$	(0, 3, 2)	108	t^{107}	$2x + 1$	(0, 2, 1)
25	t^{24}	$3x^2 + 2x + 4$	(3, 2, 4)	67	t^{66}	$3x^2 + 2x$	(3, 2, 0)	109	t^{108}	$2x^2 + x$	(2, 1, 0)
26	t^{25}	$2x^2 + 4$	(2, 0, 4)	68	t^{67}	$2x^2 + x + 4$	(2, 1, 4)	110	t^{109}	$x^2 + 4x + 1$	(1, 4, 1)
27	t^{26}	$3x + 1$	(0, 3, 1)	69	t^{68}	$x^2 + 3x + 1$	(1, 3, 1)	111	t^{110}	$4x^2 + 3x + 3$	(4, 3, 3)
28	t^{27}	$3x^2 + x$	(3, 1, 0)	70	t^{69}	$3x^2 + 3x + 3$	(3, 3, 3)	112	t^{111}	$3x^2 + x + 2$	(3, 1, 2)
29	t^{28}	$x^2 + x + 4$	(1, 1, 4)	71	t^{70}	$3x^2 + 4x + 4$	(3, 4, 4)	113	t^{112}	$x^2 + 3x + 4$	(1, 3, 4)
30	t^{29}	$x^2 + x + 3$	(1, 1, 3)	72	t^{71}	$4x^2 + 4$	(4, 0, 4)	114	t^{113}	$3x^2 + x + 3$	(3, 1, 3)
31	t^{30}	$x^2 + 3$	(1, 0, 3)	73	t^{72}	$2x + 2$	(0, 2, 2)	115	t^{114}	$x^2 + 4x + 4$	(1, 4, 4)
32	t^{31}	3	(0, 0, 3)	74	t^{73}	$2x^2 + 2x$	(2, 2, 0)	116	t^{115}	$4x^2 + x + 3$	(4, 1, 3)
33	t^{32}	$3x$	(0, 3, 0)	75	t^{74}	$2x^2 + 4x + 1$	(2, 4, 1)	117	t^{116}	$x^2 + x + 2$	(1, 1, 2)
34	t^{33}	$3x^2$	(3, 0, 0)	76	t^{75}	$4x^2 + 1$	(4, 0, 1)	118	t^{117}	$x^2 + 4x + 3$	(1, 4, 3)
35	t^{34}	$x + 4$	(0, 1, 4)	77	t^{76}	$4x + 2$	(0, 4, 2)	119	t^{118}	$4x^2 + 3$	(4, 0, 3)
36	t^{35}	$x^2 + 4x$	(1, 4, 0)	78	t^{77}	$4x^2 + 2x$	(4, 2, 0)	120	t^{119}	$x + 2$	(0, 1, 2)
37	t^{36}	$4x^2 + 2x + 3$	(4, 2, 3)	79	t^{78}	$2x^2 + 3x + 2$	(2, 3, 2)	121	t^{120}	$x^2 + 2x$	(1, 2, 0)
38	t^{37}	$2x^2 + x + 2$	(2, 1, 2)	80	t^{79}	$3x^2 + x + 1$	(3, 1, 1)	122	t^{121}	$2x^2 + 2x + 3$	(2, 2, 3)
39	t^{38}	$x^2 + x + 1$	(1, 1, 1)	81	t^{80}	$x^2 + 2x + 4$	(1, 2, 4)	123	t^{122}	$2x^2 + 2x + 1$	(2, 2, 1)
40	t^{39}	$x^2 + 3x + 3$	(1, 3, 3)	82	t^{81}	$2x^2 + x + 3$	(2, 1, 3)	124	t^{123}	$2x^2 + 1$	(2, 0, 1)
41	t^{40}	$3x^2 + 3$	(3, 0, 3)	83	t^{82}	$x^2 + 2x + 1$	(1, 2, 1)				

Table 17: Elements of $GF(5^3)$, with generating element $t = x$ and primitive polynomial $x^3 + 3x + 2$